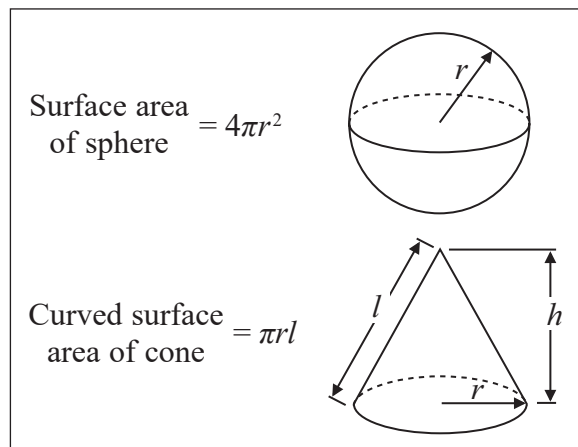
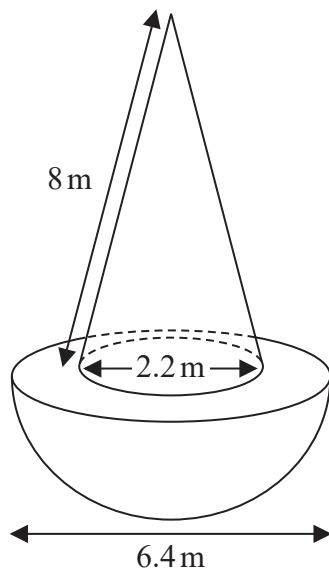


- 18 The centre of the base of a solid cone is placed at the centre of the flat face of a hemisphere to make a sculpture.



The diameter of the hemisphere is 6.4 m.

The diameter of the base of the cone is 2.2 m and the slant height of the cone is 8 m.

Callum is going to cover this sculpture with one coat of paint.

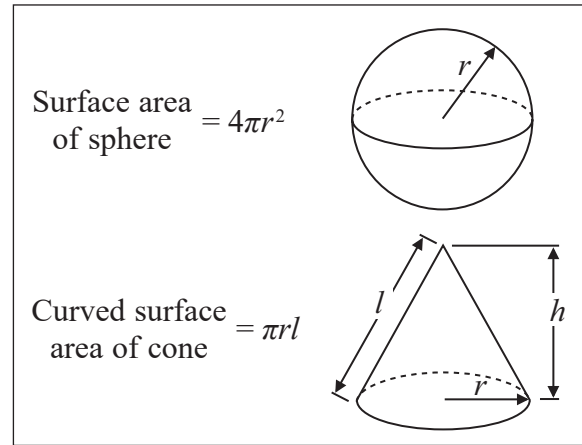
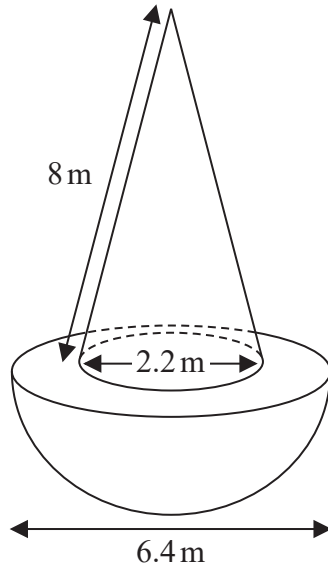
He assumes that each tin of paint will cover 8.5 m^2

Callum thinks he will need to buy 15 tins of paint.

- (a) Will 15 tins of paint be enough?

You must show how you get your answer.

- 18 The centre of the base of a solid cone is placed at the centre of the flat face of a hemisphere to make a sculpture.



The diameter of the hemisphere is 6.4 m.

The diameter of the base of the cone is 2.2 m and the slant height of the cone is 8 m.

Callum is going to cover this sculpture with one coat of paint.

He assumes that each tin of paint will cover 8.5 m^2

Callum thinks he will need to buy 15 tins of paint.

Callum finds out that each tin of paint will actually cover 7.5 m^2

(b) How does this affect your answer to part (a)?

(1)

(Total for Question 18 is 6 marks)